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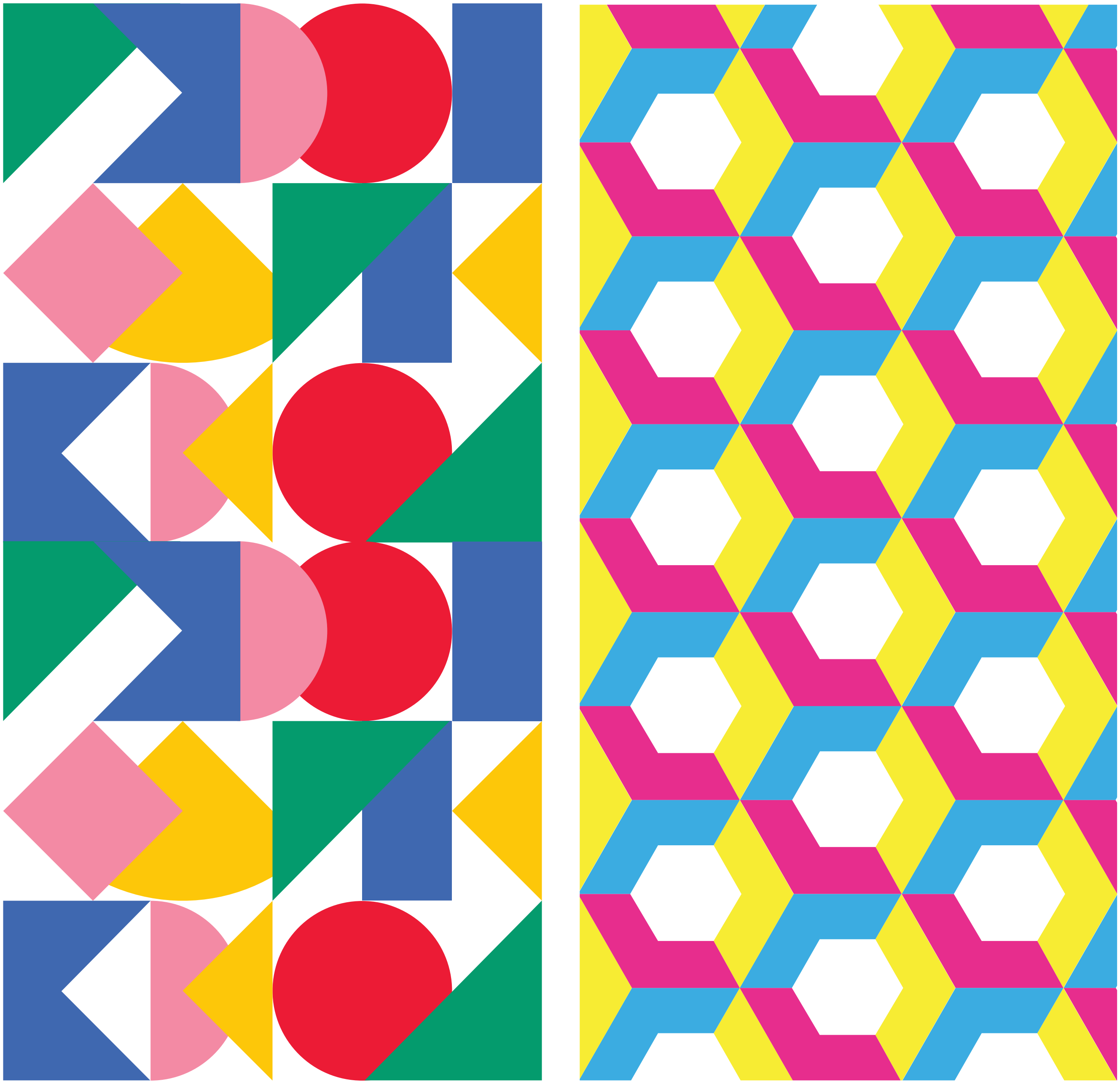

PROCEEDINGS OF CONSTRUCTIONISM / FABLEARN 2023

Edited by
Nathan Holbert &
Paulo Blikstein

CONSTRUCTIONISM

FABLEARN

The Association for Computing Machinery
1601 Broadway, 10th Floor
New York, New York 10019, USA

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INTRODUCTION

The theme for Constructionism / FabLearn 2023 was *"Building the Future of Constructionism."* While the Logo programming language was first developed in 1967, and the British Logo Group first gathered in 1983, in the years since the constructionist community has continued to evolve and expand. A critical effort of this year's conference was not just the reflection and elevation of our shared history and values, but also an explicit effort to together envision a future for where constructionism might go. Our keynotes, panels, paper presentations, and workshops told stories of who we are, what we do, how we do it, and what comes next.

At the closing panel of the conference, the co-chairs invited the attending community to critically examine the conference and where we were as a community. In reading these reactions, we considered that they would be the best way to frame these proceedings since they captured in real time what participants were excited and concerned about. Below, we have thus collated some of the conversations that occurred in that session, and used them as starting points to briefly reflect on the work presented at the conference (text in italics was shared by conference attendees.)

1. There are tensions between strengthening our shared core values and explicitly making connections (connections that have always been present) between the constructionist community and other progressive education movements.

> *Constructionism was EuroLogo. Some people decided that it was too narrow, refocused the community, and lost focus, and actual participants in the conference/community. Now, there seems to be some effort to further marginalize the community by severing even more connections to its roots.*
>
> *Are we as a community too insular - not taking into account the changing times around us? Outside, who knows or cares about constructionism?*
>
> *There is a disconnect between the constructionism community and other progressive educational movements and traditions.*
>
> *Perhaps there needs to be a "constructionism boot camp" at future events so there is a common language and folks just don't appropriate the term for their own purposes.*
>
> *Would not a more productive activity be creating a Constructionism manifesto and action plan, perhaps like the UN Sustainable Development Goals?*
>
> *We live in the era of administration, accountability, poverty, inequality, and fear of world crises. How can we communicate the value of constructionism?*

An explicit tension exists in our community between those who desire to expand the tent and those who aim to strengthen the community's existing bonds. Both aims are no doubt born out of a love of constructionism. On the one hand, it is appropriate and correct for us as a community to reflect on our history and shared values. Those values should be celebrated as revolutionary! Many of the most successful educational innovations today—from coding in schools to makerspaces—were borne out of constructionism. And though the rest of the world has taken up many constructionist ideas without naming our community, we should claim "victory." On the other hand, we desire to continue impacting the world, which is in dire need of a vision of education that values humanity and treats young people as having perspective and passion that matters. Doing so may require expanding our tent to welcome like-minded educators, scholars, and technologists. How can we do this while holding true to our core values and not excising our history and traditions? In true constructionist fashion, we believe there will never be a final answer to that dilemma but a constant conversation within our community. Out of that dialogue, we might construct not one but multiple models of how to go about acknowledging our history and principles while reinventing ourselves.

2. Making isn't new. What can we learn by paying attention to the many diverse forms of making that existed long before this educational movement was named? How might emerging fields of making offer opportunities to connect to every day and/or craft-making practices?

> *Making and learning through making have existed forever. How do we revisit ways learning through making might have looked like prior to "us" giving it a name?*
>
> *Is biomaking a way to reconnect to lost intergenerational funds of knowledge? Making yogurt, baking bread? Does the element of increased time needed for biomaking draw on gains from apprenticeship models of learning?*
>
> *What might be missing from how we talk about and study making? Making in community. Making in cultural and social representations. How would we rename constructionism? Changing the "object to think with" to something that is a "community to think with." How do we centralize community? Changing the basis of problem-solving from an individualistic base to a community base. Making for what purpose? How are the things that we are making representative of various purposes*

One would be hard-pressed to find writings about constructionism that didn't involve "making" of some kind—consider that Jeanne Bamberger's "Laboratory for Making Things" is from 1985! Papert himself credits his observation of a group of children carving soap sculptures as one of his inspirations for the foundational work of this community. Indeed, making is not new. What can we learn about making by studying making practices throughout history and across cultures? And how can we evolve constructionism by attending to these making practices and the sociocultural ecologies that brought them into being? Fundamentally, what exactly is the object of our study and interest within the very human act of making things?

3. A greater focus on schools

> *How much of constructionism's lack of presence in schools is related to research methods (quantitative vs. qualitative)? Unfortunately, grades are an effective way to collect data and used to justify a system.*
>
> *Why is there so little interest or effort being dedicated to building and strengthening connections with K-12 schools, systems, and organizations?*

"Schools are where the children are." The constructionist community has a long history of working in and directly with schools. For decades, we have built technologies, computer languages, and learning environments for classroom use. We have countless examples, the most visible today being the Scratch programming language. However, schools have also always presented challenges for the constructionist intent on revolutionizing the learning experience. While teachers and students are often willing to transform the classroom, schools, as large systems, are often resistant to change.

To what extent should we focus on evolution (slow, gradual change) vs. revolution (radical shifts)? There is wide agreement in the community that our efforts must include schools.
In some cases, that may mean adapting our revolutionary designs to accommodate the constraints of the classroom. Doing so doesn't have to compromise our core commitments. We have seen schools change! Computer science for all and makerspaces are two recent examples. However, we should not be content with incremental change and should never abandon our commitment to revolution.

We believe these proceedings of Constructionism / Fablearn 2023 represent our community well. In them, we see our community's deep commitment to children and humanity; scholarship that leverages the tools of scientific inquiry to understand vital questions of learning, teaching, and design; and a passion for making our world a more just and equitable place.

Like a family, we never quite agree on everything. We argue and get frustrated with one another, we refuse absolute truths or infallible leaders. And still, year after year, the conference transpires a deep sense of love and respect for each other. Constructionists have been predicting their demise for decades; nevertheless, here we are, refusing to go away and becoming more and more relevant as the years go by. We are honored to be part of this community and look forward to working with you all to continue building a bright future for constructionism and what it represents.

We are not going anywhere but forward (50).

Nathan Holbert & Paulo Blikstein
Editors, Proceedings of Constructionism / FabLearn 2023

This volume includes full and short research papers published by ACM. Another volume of works presented at the conference, published by ETC Press, can be found at https://press.etc.cmu.edu/proceedings/proceedings-constructionism-fablearn-2023

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PaperCut: A Craft-Centered Approach to Digital Fabrication Using Cut Paper as a Design Medium

Amy Traylor
traylоramy@unm.edu
University of New Mexico
Department of Language, Literacy, and Sociocultural Studies
Albuquerque, New Mexico, USA

Ruby Ta
nta@unm.edu
University of New Mexico
Department of Computer Science
Albuquerque, New Mexico, USA

Alyssa Johnson
ajohnson87@unm.edu
University of New Mexico
Department of Fine Arts
Albuquerque, New Mexico, USA

Leah Buechley
buechley@unm.edu
University of New Mexico
Department of Computer Science
Albuquerque, New Mexico, USA

Figure 1: An example of a student's PaperCut project. Paper cutouts define the form and surface texture of a vessel, which is visualized in the software and then 3D printed in play-dough.

ABSTRACT

This paper introduces PaperCut, software that enables novices to design textured 3D vessels from paper cutouts. The software and accompanying workflow were developed to enable novices to explore 3D printing entirely through hands-on craft-based activities, bypassing traditional Computer Aided Design (CAD) and Computer Aided Machining (CAM) software workflows. The approach is designed to enable novice creators to apply their expertise in traditional craft to interactions with a 3D printer. We describe our approach and discuss a workshop in which participants used PaperCut to create 3D printed forms in clay and a custom 3D printable play-dough.

CCS CONCEPTS

• **Human-centered computing → Interactive systems and tools**.

KEYWORDS

digital fabrication, craft, 3D printing, play-dough, clay, paper cutouts

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1 INTRODUCTION

In a traditional approach to 3D printing, a designer first constructs a 3D model using Computer Aided Design (CAD) software like Rhino, Blender, TinkerCad, or BeetleBlocks. Then, the model is prepared for printing using Computer Aided Machining (CAM) slicing software like Cura. Slicing software generates a toolpath–instructions in the .gcode file format–which the 3D printer follows to produce the part. This workflow requires a designer to learn and work with complex software programs that are divorced from craft and physical making [10]. The workflow presents significant hurdles to many, including educators and young people, who may be interested in engaging with 3D printing but lack expertise in CAD and CAM software.

Researchers have explored a variety of ways to integrate physical craft experiences into making activities [3, 6, 7]. A related body of work has investigated ways to make digital fabrication more approachable to novices. Software like BeetleBlocks [9], TinkerCad, and CodeBlocks [11] attempt to make the process of 3D modeling simple and intuitive for young people and novices. Other software has been developed for specific fabrication machines. Turtlestitch [12] and PEMbroider [2] are tools that support computational design for embroidery. Clayon [1] and Potterware [8] are tools that support the design of vessels for clay 3D printing. These tools, which enable users to create vase-like forms with a range of surface textures provide some of the same design opportunities that PaperCut does, are much more traditional CAD oriented software.

Our approach is distinct in employing paper–a physical craft material–as the primary design medium for digital fabrication. Our intent is to allow designers to work in the physical world as much as possible. Software is deemphasized, used to translate a design from one physical craft medium (paper) into another (clay or play-dough). We see this work as contributing an example of how we can enrich and expand the landscape of 3D printing for children, responding to Eisenberg's call for a richer and more materially-complex set of opportunities in this domain [5].

PaperCut is software that enables people to design 3D prints by collaging paper cutouts. Our software can generate designs suitable for a range of 3D printers including traditional Fused Filament Fabrication machines that print in PLA. Here we emphasise the connection to craft, by focusing on clay and play-dough 3D printing.

The primary contributions of this work are: 1) The creation of software and workflows that support approaches to 3D printing which bypass traditional modeling software and emphasize craft materials like paper. 2) The results from a workshop that documents the useability of our software. 3) An identification of opportunities for ongoing work in craft-centered design approaches to make digital fabrication more accessible.

2 SOFTWARE AND WORKFLOW

To begin a project, a user creates a set of paper cut outs. The form of a vessel is defined by attaching a shape cut out of white paper to a black background. The white form creates the basic silhouette of the vessel. A second cutout, made from gray and white paper on a black background, defines the vessel's surface texture, with white and grey corresponding to different indentation depths in the surface.

Once a set of paper cutouts has been created, these are scanned and loaded into the PaperCut software. The software can accommodate up to six cutouts each for form and texture, see Figure 2. The user chooses one form and one texture cutout and clicks "REDRAW" to get a preview of the three-dimensional form in the main window.

Each form is associated with a maximum radius and a minimum radius. These values determine how narrow or wide the final vessel will be, while maintaining the basic shape defined by the cutout.

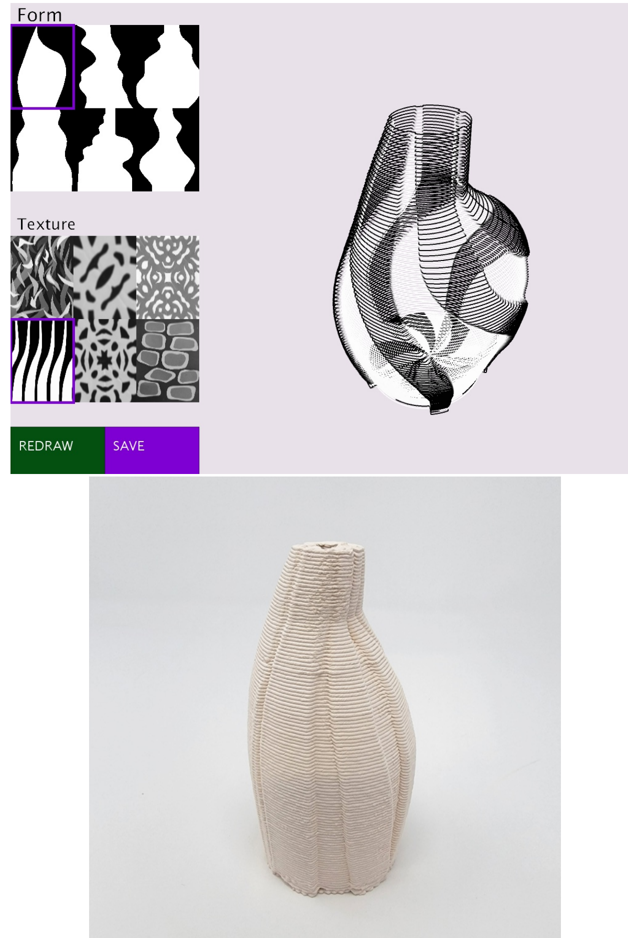

Figure 2: The PaperCut software showing a design and corresponding 3D print in clay.

Narrower vessels, in which there is a relatively small difference between the maximum and minimum radius values, are more likely to result in successful 3D prints, as they avoid the creation of unsupported overhangs. The depth of the surface textures on a form are controlled with a similar set of variables.

Once the user is satisfied with a design, they can click the "SAVE" button to export an image of the current interface settings along with a printable .gcode file. The .gcode file is sent to a printer which generates the final artifact. All prints described in this paper were made on an Eazao Zero, a robust and affordable (under $1000 USD), clay 3D printer. Prints were created from clay or a custom 3D printable play-dough–we use a recipe developed by Buechley et al. [4].

3 WORKSHOP

To demonstrate the viability of our approach and elicit feedback, we conducted a four-hour workshop with 19 children ages 8-14, 9 young women, 8 young men and 1 non-binary youth. We began with a general introduction to 3D printing and the workshop goals. Students then created paper cutouts for form and surface texture, each making at least two distinct forms and textures. Workshop leaders helped students scan and transfer these designs into the software. Students adjusted their forms and generated .gcode files which were printed on an Eazao Zero, Figure 3 left.

Each print was approximately 4 x 1.5 inches (100 x 40mm) and took approximately 15 minutes to print. We had four printers running during the workshop. Each was loaded with different colors of play-dough. Students were able to choose which printer and corresponding play-dough color they wanted to use. Once prints were complete, they were transferred to a dehydrator where they were left for approximately one hour for drying before they could be taken home. The play-dough prints are solid and stable, but cannot hold water unless they are sealed with a waterproof coating.

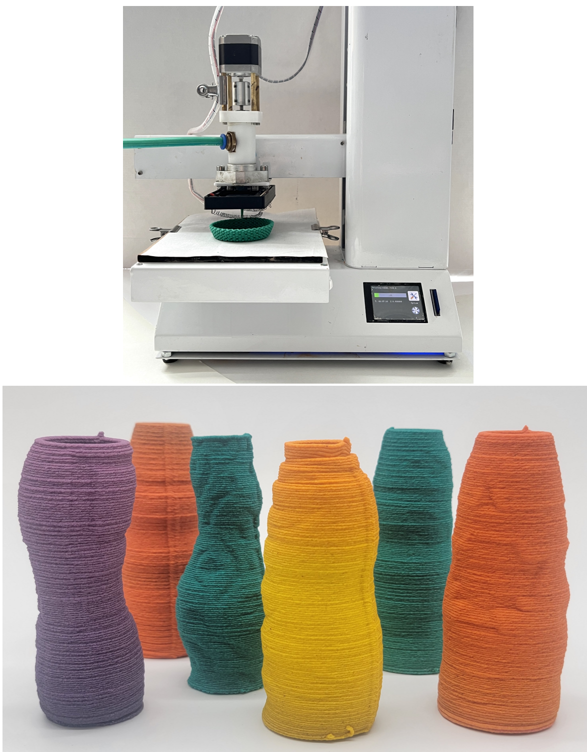

Figure 3: Left: The Eazao Zero 3D printer printing a design in play-dough. Right: A collection of vessels made in our workshop.

All workshop participants were able to create cut paper patterns for both form and texture. A few of the youth participants created some designs that were not printable. One student, for instance, created an elaborate dragon cutout that could not be transformed into a vessel. All participants were ultimately able to design and create at least one printable form. Figure 3 right shows a collection of designs created by youth in our workshop.

We believe that this preliminary exploration demonstrates that designing with cut paper is a viable path to generating 3D form. The enthusiasm of the novice fabricators for the workflow, the machine process, and the final forms shows the possibilities of new entrants into the field of fabrication.

Some limitations of the software was in the scanning of the 3D designs from cut paper. Stopping to scan the designs created a bottleneck that stalled the progress of the workflow from design to software to printer. Future iterations of software and workflow will need to address the issue of scanning and editing designs for use in the software. We also plan to include more control features and feedback mechanisms in our interface to help users understand the variables that determine the shape and textures of their vessels before they are printed. We would also like to hold longer sessions with fewer students so that students are able to work through more than one design iteration. We believe that being able to design and print more than one vessel would increase the range of designs students were able to create and help us better understand the creative affordances of our approach.

4 CONCLUSION

We believe there is ample space in the research on digital fabrication for the exploration of novel methods to design for and with different materials and machines. We believe that approaches like the one we have described, that are anchored in physical arts and craft activities, could lead to a broader audience's exploration of 3D printing. The emphasis on physical materials in the design process shows that we can alleviate the need to always use general purpose modeling software and slicers. An approach like ours–facilitated by very light-weight software–could potentially be implemented on a phone or tablet. We believe that returning to the power of physical design material's and tool's ability to influence digital space are worthy avenues of speculation.

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